Robust Secure UAV Communication Systems with Full-Duplex Jamming

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Abstract—**In this paper, we study the robust secure unmanned aerial vehicle (UAV) communication system, where a UAV with full-duplex (FD) capability simultaneously receives the information signal from a ground unit (GU) and transmits jamming signal to degrade the wiretap capability of potential multiple ground eavesdroppers (Eves). With the consideration of estimation error of Eves' locations, we aim to maximize the average worst secrecy rate inside a certain flight period of the UAV by jointly optimizing the transmit power of the GU and UAV as well as the trajectory of the UAV. The resulting problem is intractable due to its non-convex nature and strongly coupled variables. Furthermore, the estimation error of Eves' locations results in an infinite number of constraints, which makes the problem even more difficult. To tackle this difficulty, we first propose an iterative algorithm based on the Schur complement lemma and successive inner approximation method to efficiently solve the problem suboptimally under the estimated Eves' locations. Then, in order to cope with the of Eves' location errors, we develop a cutting-set method, which solves the problem by alternating between optimal power-trajectory design and worst-case Eves' locations analysis. Via simulation, we show the improvement of the proposed algorithm compared to other benchmark algorithms under high FD self-interference cancellation levels.**

Index Terms—**Secure UAV communication, robust design, full-duplex, jamming.**

I. Introduction

Thanks to high mobility, low cost and quick deployment, unmanned aerial vehicles (UAVs) have emerged in recent years as a new type of aerial transceiver (e.g., as a user, base station or relay) in wireless communication, and are envisioned to play an important role in the forthcoming fifth-generation (5G) [1], [2]. Despite various UAV aided applications, UAV communication is more prone to be intercepted by ground eavesdroppers (Eves) due to the general existence of line-of-sight (LoS) air-to-ground wireless links, showing a significant security challenge in UAV wireless communication. On the other hand, taking advantage of flexible mobility of the UAV, physical layer security is shown to enhance the secure UAV communication by leveraging proper trajectory design [3].

Crucial components in secure UAV communication are transmit power control and trajectory design. In [4], by jointly optimizing the power allocation and UAV trajectory, the average secrecy rate is improved for a UAV communication system with a single eavesdropper (Eve) whose location is perfectly known. Furthermore, artificial jamming can usually bring a significant enhancement to the physical layer security system. In [5] the UAV enabled cooperative jamming was investigated to enhance the secrecy rate of ground wiretap channels. The authors in [6] considered a dual UAV scenario where one UAV communicates with legitimate ground users while the other UAV sends cooperative jamming. Although the imperfect location information of Eves was considered in [6], the authors did not provide a robust design. Only the lower bound of the secrecy rate was optimized, which could be very different from the real secrecy rate when the uncertain region of Eve's location is large. By introducing full-duplex (FD) capability to the UAV, the authors in [7] proposed a secure UAV communication system with FD jamming. However, this work assumed the perfect knowledge of Eve's location, which is overly optimistic in practice. This assumption of perfect location information may lead to a large performance loss especially in the case with multiple Eves [8], which was also not considered in [7].

Motivated by the above, we study a secure UAV communication system, where FD jamming of UAV is exploited to maximize the average worst secrecy rate under multiple eavesdroppers whose locations are not perfectly known. The main contributions are: i) For a scenario with given Eves' locations, by applying the Schur complement lemma and successive inner approximation (SIC) method [9], we transform the intractable original problem with strongly coupled variables into a tractable iteratively solvable problem, where in each iteration a semidefinite programming (SDP) with linear matrix inequalities (LMIs) is solved. Note that different from the aforementioned works that they all apply the block coordinate descent method to alternately optimize the prime variables, our algorithm optimizes all prime variables jointly to avoid bad local optimum. ii) To cope with the uncertainty of Eves' locations, we propose a cutting-set method [10] that solves the problem by alternating between an optimization step and a pessimization step. In the optimization step, the transmit power and the 3D-trajectory of the UAV are optimally designed under a given finite subset of the uncertainty region of Eves' locations. In the pessimization step, the subset is updated through the computation of the worst-case locations in uncertainty regions.

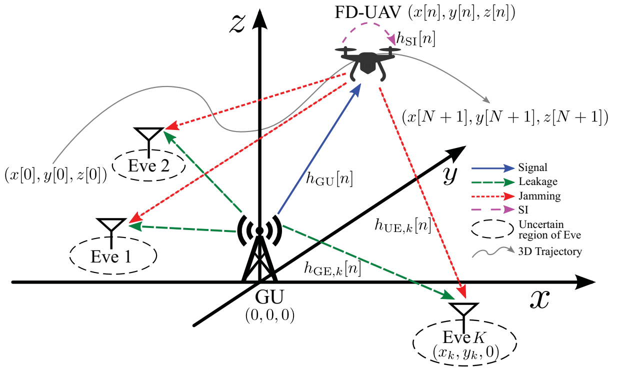

Fig. 1. An illustration of studied FD-UAV secure communication system.

II. System Model

We study a UAV aided secure wireless communication system, where one ground units (GU) transmits the information signal to the UAV, meanwhile K potential ground Eves are wiretapping the information, as shown in Fig. 1. To degrade the capability of the wiretap channels, the UAV with FD capability transmits jamming signal to the potential Eves while it receives the information signal from the GU. We assume that all nodes are equipped with single antenna. Considering a three dimensional Cartesian coordinate system, without loss of generality we assume that the location of the GU is (0, 0, 0) and denote the location of the k-th Eve as $\mathbf{c}_k = (x_k, y_k, 0)$, where $k \in \mathcal{K} = \{1, \ldots, K\}$. We consider the imperfect location information of Eves. Specifically, we do not know the exact locations of Eves, i.e., $\mathbf{c}_k, \forall k \in \mathcal{K}$ are unknown. We only know an estimated location of the k-th Eve, denoted as $\hat{\mathbf{c}}_k = (\hat{x}_k, \hat{y}_k, 0)$ with the estimation error, denoted as $\Delta\mathbf{c}_k = (\Delta x_k, \Delta y_k, 0)$, where $\mathbf{c}_k = \hat{\mathbf{c}}_k + \Delta\mathbf{c}_k$. We assume that the estimation error follows the constraint $\Delta\mathbf{c}_k \in \Omega_k \triangleq \{\|\Delta\mathbf{c}_k\| \leq \epsilon_k\}, \forall k \in \mathcal{K}$, where $\epsilon_k \in \mathcal{R}^+$ is the bound of the norm-2 estimation error of the k-th Eve location.

We consider that the UAV needs to finish a specific task with a total flying time of T_{tot}. For easy analyzing, T_{tot} is divided into N time slots, where the duration of each time slot is $\delta = T_{\mathrm{tot}}/N$[1]. Within the given flying time, the UAV should fly from the start point, denoted as $\mathbf{q}[0] = (x[0], y[0], z[0])$, to the end point, denoted as $\mathbf{q}[N+1] = (x[N+1], y[N+1], z[N+1])$, and communicate to the GU. Moreover, at the n-th time slot the location of the UAV is denoted as $\mathbf{q}[n] = (x[n], y[n], z[n])$, where $n \in \mathcal{N} = \{1, \ldots, N\}$. Thus, at the n-th time slot the horizontal speed of the UAV is denoted as $V[n] = \sqrt{(x[n+1] - x[n])^2 + (y[n+1] - y[n])^2}/\delta$. For practical safety regulations, the flying process of the UAV should satisfy the following constraints:

$$z_{\min} \leq z[n] \leq z_{\max},\ V[n] \leq V_{\max},\ \forall n \in \{0\} \cup \mathcal{N}, \tag{1a}$$

$$z[n+1] - z[n] \leq V_{\mathrm{up}}\delta,\ \forall n \in \{0\} \cup \mathcal{N}, \tag{1b}$$

$$z[n] - z[n+1] \leq V_{\mathrm{down}}\delta,\ \forall n \in \{0\} \cup \mathcal{N}, \tag{1c}$$

[1]Note that δ should be sufficiently small so that the location of the UAV can be assumed to be unchanged within every time slot.

where $z_{\max}$ and $z_{\min}$ are maximum and minimum allowed altitudes of the UAV, respectively. Moreover, $V_{\max}$, V_{up} and V_{down} are the maximum horizontal, vertical ascending and vertical descending speed, respectively.

A. Signal model

At n-th time slot the information and jamming signal of the GU and UAV are respectively represented as $s[n] \sim \mathcal{CN}(0, P_{\mathrm{G}}[n])$ and $w[n] \sim \mathcal{CN}(0, P_{\mathrm{U}}[n])$, where $P_{\mathrm{G}}[n]$ and $P_{\mathrm{U}}[n]$ are the transmit and jamming power of the GU and UAV at the n-th time slot, respectively. Then, the received signals of the UAV and the k-th Eve at the n-th time slot are respectively given as

$$y_{\mathrm{U}}[n] = h_{\mathrm{GU}}[n]s[n] + h_{\mathrm{SI}}[n]w[n] + n_{\mathrm{U}}[n], \tag{2}$$

$$y_{\mathrm{E},k}[n] = h_{\mathrm{GE},k}[n]s[n] + h_{\mathrm{UE},k}[n]w[n] + n_{\mathrm{E},k}[n], \tag{3}$$

where $h_{\mathrm{GU}}[n], h_{\mathrm{GE},k}[n]$ and $h_{\mathrm{UE},k}[n]$ are the channels at the n-th time slot between GU and UAV, GU and the k-th Eve as well as UAV and the k-th Eve, respectively. Moreover, due to the imperfect self-interference (SI) cancellation of the FD transceiver, the UAV suffers from its own transmit signal through the SI channel denoted as $h_{\mathrm{SI}}[n]$ at the n-th time slot. Furthermore, $n_{\mathrm{U}}[n] \sim \mathcal{CN}(0, N_{\mathrm{U}})$ and $n_{\mathrm{E},k}[n] \sim \mathcal{CN}(0, N_{\mathrm{E},k})$ are the additive white noise at the receiver antennas of UAV and the k-th Eve, respectively. We assume that the noise power of UAV and Eves remain the same during T_{tot} as N_{U} and $N_{\mathrm{E},k}, \forall k \in \mathcal{K}$.

For the practical consideration, we denote the peak and average power of the GU as $P_{\mathrm{G}}^{\mathrm{peak}}$ and $P_{\mathrm{G}}^{\mathrm{avg}}$, and of the UAV as $P_{\mathrm{U}}^{\mathrm{peak}}$ and $P_{\mathrm{U}}^{\mathrm{avg}}$, respectively. Then, the power should follow the constraints as

$$0 \leq P_{\mathrm{G}}[n] \leq P_{\mathrm{G}}^{\mathrm{peak}}, \quad 0 \leq P_{\mathrm{U}}[n] \leq P_{\mathrm{U}}^{\mathrm{peak}},\ \forall n \in \mathcal{N}, \tag{4a}$$

$$\frac{1}{N}\sum_{n=1}^{N} P_{\mathrm{G}}[n] \leq P_{\mathrm{G}}^{\mathrm{avg}}, \quad \frac{1}{N}\sum_{n=1}^{N} P_{\mathrm{U}}[n] \leq P_{\mathrm{U}}^{\mathrm{avg}}, \tag{4b}$$

where for the concern of non-trivial constraints, we assume $P_{\mathrm{G}}^{\mathrm{avg}} \leq P_{\mathrm{G}}^{\mathrm{peak}}$ and $P_{\mathrm{U}}^{\mathrm{avg}} \leq P_{\mathrm{U}}^{\mathrm{peak}}$.

B. Channel model

We assume that all air-ground channels are LoS-dominated. Then, we adopt the free-space path loss model to the air-ground channels. Thus, the channel power gains of the GU to the UAV channel and the UAV to the k-th Eve channel at the n-th time slot are given as $|h_{\mathrm{GU}}[n]|^2 = \beta_0/\|\mathbf{q}[n]\|^2$ and $|h_{\mathrm{UE},k}[n]|^2 = \beta_0/\|\mathbf{c}_k - \mathbf{q}[n]\|^2$, respectively, where β_0 presents the power gain of the channel at the reference distance of unit meter. The ground-to-ground channels between the GU and Eves are assumed to follow Rayleigh fading. Thus, the power gain of the channel between the GU and the k-th Eve is given as $|h_{\mathrm{GE},k}[n]|^2 = \bar{h}_{\mathrm{GE},k}\zeta$, where $\bar{h}_{\mathrm{GE},k} = \beta_0/\|\mathbf{c}_k\|^{\alpha}$ presents the path-loss with $\alpha \geq 2$ as the path-loss exponent, and ζ is an exponentially distributed random variable with unit mean presenting the small-scale Rayleigh fading. Moreover, we consider the residual SI of the FD radio in UAV due to imperfect loop interference cancellation. The SI channel is modeled as Rayleigh fading with $\mathbb{E}(|h_{\rm SI}[n]|^2) = \sigma_{\rm SI}^2$, where $\sigma_{\rm SI}^2$ is regarded as the average loop interference level [11].

C. Problem formulation

With the received signals $y_{\rm U}[n]$ and $y_{\rm E,k}[n]$, the resulting channel capacities of the channels between the GU and UAV as well as the GU and the k-th Eve at the n-th time slot are respectively given as

$$\begin{aligned} R_{\rm U}[n] &= \mathbb{E}_{|h_{\rm SI}[n]|^2}\left\{\log_2\left(1+\frac{|h_{\rm GU}[n]|^2 P_{\rm G}[n]}{|h_{\rm SI}[n]|^2 P_{\rm U}[n]+N_{\rm U}}\right)\right\} \\ &\overset{(a)}{\geq} \log_2\left(1+\frac{|h_{\rm GU}[n]|^2 P_{\rm G}[n]}{\sigma_{\rm SI}^2 P_{\rm U}[n]+N_{\rm U}}\right) \\ &\triangleq \check{R}_{\rm U}[n], \end{aligned} \tag{5}$$

$$\begin{aligned} R_{{\rm E},k}[n] &= \mathbb{E}_{\zeta}\left\{\log_2\left(1+\frac{|h_{{\rm GE},k}[n]|^2 P_{\rm G}[n]}{|h_{{\rm UE},k}[n]|^2 P_{\rm U}[n]+N_{{\rm E},k}}\right)\right\} \\ &\overset{(b)}{\leq} \log_2\left(1+\frac{\bar{h}_{{\rm GE},k} P_{\rm G}[n]}{|h_{{\rm UE},k}[n]|^2 P_{\rm U}[n]+N_{{\rm E},k}}\right) \\ &\triangleq \hat{R}_{{\rm E},k}[n], \end{aligned} \tag{6}$$

where $\mathbb{E}_{|h_{\rm SI}[n]|^2}$ and $\mathbb{E}_{\zeta}$ are the expectation operators with respect to $|h_{\rm SI}[n]|^2$ and ζ, respectively. (a) in (5) and (b) in (6) hold based on the Jensen's inequality due to the convexity of $R_{\rm U}[n]$ and concavity of $R_{{\rm E},k}[n]$ with respect to the corresponding random variables. Thus, the worst achievable secrecy rate among all Eves and the GU in the n-th time slot is lower bounded by [12], [13]

$$R_{\rm sec}[n] = \left\{\check{R}_{\rm U}[n] - \max_{k\in\mathcal{K}} \max_{\Delta\mathbf{c}_k\in\Omega_k} \hat{R}_{{\rm E},k}[n]\right\}^+, \tag{7}$$

where $\{x\}^+ = \max(x, 0)$. We aim to maximize the worst average secrecy rate among all Eves, which results in the following optimization problem

$$\begin{aligned} \max_{\mathbb{P}_{\rm G},\mathbb{P}_{\rm U},\mathbb{Q}} \quad & \frac{1}{N}\sum_{n=1}^{N} R_{\rm sec}[n] \\ \text{s.t.} \quad & (1),\ (4), \end{aligned} \tag{8}$$

where $\mathbb{P}_{\rm G} = \{P_{\rm G}[n]\}$, $\mathbb{P}_{\rm U} = \{P_{\rm U}[n]\}$ and $\mathbb{Q} = \{\mathbf{q}[n]\}$ are the variable sets. It is noticed that the optimization problem in (8) is intractable due to many facts. We first cope with the non-linear operator $\{\cdot\}^+$ in (7). Fortunately, the non-linear operator $\{\cdot\}^+$ at the optimality of problem (8) has no influence since the GU at n-th time slot can transmit no signal to avoid a negative $R_{\rm sec}[n]$. As a result, we can ignore this operator during the optimization. Then, by introducing auxiliary variables $\mathbb{R} = \{R_{\rm E}^{\max}[n]\}$, the problem in (8) can be equivalently reformulated as

$$\max_{\mathbb{P}_{\rm G},\mathbb{P}_{\rm U},\mathbb{Q},\mathbb{R}} \quad \frac{1}{N}\sum_{n=1}^{N}\left(\check{R}_{\rm U}[n] - R_{\rm E}^{\max}[n]\right) \tag{9a}$$

$$\text{s.t.} \quad R_{\rm E}^{\max}[n] \geq \max_{\Delta\mathbf{c}_k\in\Omega_k} \hat{R}_{{\rm E},k}[n],\ \forall k\in\mathcal{K}, n\in\mathcal{N}, \tag{9b}$$

$$(1),\ (4). \tag{9c}$$

Note that the optimization problem in (9) is still intractable due to the fact that the problem is not jointly convex over the variables $\mathbb{P}_{\rm G}, \mathbb{P}_{\rm U}$ and $\mathbb{Q}$. Moreover, the estimation error of the Eves' locations implies an infinite number of constraints which makes the problem even more difficult to solve. In the next section, we first propose an efficient solution based on the Schur complement lemma and successive inner approximation method to iteratively solve the problem in (9) without considering the uncertainty of Eves' locations. Then, we adopt the cutting-set method to cope with the estimation error of the Eves' locations in Section IV.

III. Joint Power and Trajectory Optimization under Known Eves' Locations

In this section, we solve the optimization in (9) without considering the uncertainty of Eves' locations. By ignoring the estimation error of the Eves' locations, i.e., let $\mathbf{c}_k := \hat{\mathbf{c}}_k, \forall k \in \mathcal{K}$, the problem in (9) can be rewritten as

$$\begin{aligned} \max_{\mathbb{P}_{\rm G},\mathbb{P}_{\rm U},\mathbb{Q},\mathbb{R}} \quad & (9a) \\ \text{s.t.} \quad & R_{\rm E}^{\max}[n] \geq \hat{R}_{{\rm E},k}[n],\ \forall k\in\mathcal{K}, n\in\mathcal{N}, \\ & (1),\ (4). \end{aligned} \tag{10}$$

For easy analysis, the problem in (10) can be reformulated as

$$\max_{\mathbb{P}_{\rm G},\mathbb{P}_{\rm U},\mathbb{Q},\mathbb{R}} \quad \frac{1}{N}\sum_{n=1}^{N}\left(R_1[n] - R_2[n] - R_{\rm E}^{\max}[n]\right) \tag{11a}$$

$$\text{s.t.} \quad R_{\rm E}^{\max}[n] \geq R_3[n] - R_4[n],\ \forall k\in\mathcal{K}, n\in\mathcal{N}, \tag{11b}$$

$$(1),\ (4), \tag{11c}$$

where for the simplicity of notation we define $\mathbf{a}_k[n] := \mathbf{q}[n] - \mathbf{c}_k$ and

$$R_1[n] := \log_2\left(\frac{\beta_0 P_{\rm G}[n]}{(\mathbf{q}[n])^{\rm T}\mathbf{q}[n]} + \sigma_{\rm SI}^2 P_{\rm U}[n] + N_{\rm U}\right), \tag{12}$$

$$R_2[n] := \log_2\left(\sigma_{\rm SI}^2 P_{\rm U}[n] + N_{\rm U}\right), \tag{13}$$

$$R_3[n] := \log_2\left(\frac{\beta_0 P_{\rm G}[n]}{(\mathbf{c}_k^{\rm T}\mathbf{c}_k)^{\frac{\alpha}{2}}} + \frac{\beta_0 P_{\rm U}[n]}{(\mathbf{a}_k[n])^{\rm T}\mathbf{a}_k[n]} + N_{{\rm E},k}\right), \tag{14}$$

$$R_4[n] := \log_2\left(\frac{\beta_0 P_{\rm U}[n]}{(\mathbf{a}_k[n])^{\rm T}\mathbf{a}_k[n]} + N_{{\rm E},k}\right). \tag{15}$$

It is noticed that the optimization problem in (11) is still intractable due to the coupled variables and non-convex terms. Specifically, the variables $P_{\rm G}[n]$ and $\mathbf{q}[n]$ in $R_1[n]$ as well as the variables $P_{\rm U}[n]$ and $\mathbf{q}[n]$ in R_3 and R_4 are in a fractional form, which is not jointly convex. Moreover, $-R_2[n]$ in the objective function (11a) is convex over variable $P_{\rm U}[n]$, which also makes the problem non-convex. In the following, we first decouple the variables with the fractional form by introducing auxiliary variables and then transform them into the LMIs using the Schur Complement Lemma. After that, we cope with the non-convex terms using the SIC method to iteratively solve the whole problem.

First, by introducing the positive auxiliary variables $\mathbb{T}=\{\{t_1[n]>0\},\{t_{2,k}[n]>0\},\{t_{3,k}[n]>0\}\}$ the problem in (11) can be equivalently reformulated as

$$\max_{\mathbb{P}_{\mathrm{G}},\mathbb{P}_{\mathrm{U}},\mathbb{Q},\mathbb{R},\mathbb{T}} \frac{1}{N}\sum_{n=1}^{N}\Big(\log_2\left(t_1^{-1}[n]+\sigma_{\mathrm{SI}}^2P_{\mathrm{U}}[n]+N_{\mathrm{U}}\right) - \log_2\left(\sigma_{\mathrm{SI}}^2P_{\mathrm{U}}[n]+N_{\mathrm{U}}\right)-R_{\mathrm{E}}^{\max}[n]\Big) \tag{16a}$$

$$\text{s.t.}\quad R_{\mathrm{E}}^{\max}[n]\geq\log_2\left(\frac{\beta_0P_{\mathrm{G}}[n]}{(\mathbf{c}_k^{\mathrm{T}}\mathbf{c}_k)^{\frac{\alpha}{2}}}+t_{2,k}[n]+N_{\mathrm{E},k}\right) -\log_2\left(t_{3,k}^{-1}[n]+N_{\mathrm{E},k}\right),\ \forall k\in\mathcal{K},n\in\mathcal{N}, \tag{16b}$$

$$t_1^{-1}[n]\leq\frac{\beta_0P_{\mathrm{G}}[n]}{(\mathbf{q}[n])^{\mathrm{T}}\mathbf{q}[n]},\ \forall n\in\mathcal{N}, \tag{16c}$$

$$t_{2,k}[n]\geq\frac{\beta_0P_{\mathrm{U}}[n]}{(\mathbf{a}_k[n])^{\mathrm{T}}\mathbf{a}_k[n]},\ \forall k\in\mathcal{K},n\in\mathcal{N}, \tag{16d}$$

$$t_{3,k}^{-1}[n]\leq\frac{\beta_0P_{\mathrm{U}}[n]}{(\mathbf{a}_k[n])^{\mathrm{T}}\mathbf{a}_k[n]},\ \forall k\in\mathcal{K},n\in\mathcal{N}, \tag{16e}$$

$$(1),\ (4). \tag{16f}$$

Then, we rewrite constraints (16c), (16d) and (16e) as finite LMIs. First, constraints (16c), (16d) and (16e) can be respectively reformulated as

$$\beta_0P_{\mathrm{G}}[n]-(\mathbf{q}[n])^{\mathrm{T}}(t_1^{-1}[n]\mathbf{I})\mathbf{q}[n]\geq 0,\ \forall n\in\mathcal{N}, \tag{17}$$

$$-\beta_0P_{\mathrm{U}}[n]+(\mathbf{a}_k[n])^{\mathrm{T}}(t_{2,k}[n]\mathbf{I})\mathbf{a}_k[n]\geq 0,\ \forall k\in\mathcal{K},n\in\mathcal{N}, \tag{18}$$

$$\beta_0P_{\mathrm{U}}[n]-(\mathbf{a}_k[n])^{\mathrm{T}}(t_{3,k}^{-1}[n]\mathbf{I})\mathbf{a}_k[n]\geq 0,\ \forall k\in\mathcal{K},n\in\mathcal{N}. \tag{19}$$

It is known that the Schur Complement Lemma states: if $A\succ 0$, then $X/A=C-B^{\mathrm{T}}A^{-1}B\succeq 0\Leftrightarrow X\succeq 0$, where

$$X=\begin{bmatrix}A & B\\ B^{\mathrm{T}} & C\end{bmatrix}. \tag{20}$$

Note that $t_1[n]>0, t_{2,k}[n]>0, t_{3,k}[n]>0, \forall n\in\mathcal{N}, k\in\mathcal{K}$. Thus, by applying the Schur Complement Lemma the constraints (17) and (19) can be equivalently stated as the LMIs

$$\begin{bmatrix}t_1[n]\mathbf{I} & \mathbf{q}[n]\\ (\mathbf{q}[n])^{\mathrm{T}} & \beta_0P_{\mathrm{G}}[n]\end{bmatrix}\succeq 0,\ \forall n\in\mathcal{N}, \tag{21}$$

$$\begin{bmatrix}t_{3,k}[n]\mathbf{I} & \mathbf{a}_k[n]\\ (\mathbf{a}_k[n])^{\mathrm{T}} & \beta_0P_{\mathrm{U}}[n]\end{bmatrix}\succeq 0,\ \forall k\in\mathcal{K},n\in\mathcal{N}. \tag{22}$$

As for the constraint (18), it may not be directly transformed into the LMIs, due to its non-convex nature. The following lemma constitutes an equivalent tractable form for the constraint (18).

Lemma 1. *Let $\{\boldsymbol{\theta}_k[n]\}$ be a set of auxiliary variables and $\gamma_k[n]:=-\beta_0P_U[n]+(\boldsymbol{\theta}_k[n])^T\mathbf{a}_k[n]+(\mathbf{a}_k[n])^T\boldsymbol{\theta}_k[n]$. Then, the constraint: $\exists\boldsymbol{\theta}_k[n]\big|$*

$$\gamma_k[n]-(\boldsymbol{\theta}_k[n])^T(t_{2,k}^{-1}[n]\mathbf{I})\boldsymbol{\theta}_k[n]\geq 0,\ \forall k\in\mathcal{K},n\in\mathcal{N}, \tag{23}$$

shares the same feasible space over the variables $P_U[n]$, $t_2[n]$ and $\mathbf{a}_k[n]$ as the constraint presented in (18).

Proof. Let $f:=\gamma_k[n]-(\boldsymbol{\theta}_k[n])^{\mathrm{T}}(t_{2,k}^{-1}[n]\mathbf{I})\boldsymbol{\theta}_k[n]$. Due to the concave quadratic nature, the $\boldsymbol{\theta}_k[n]$ corresponding to the maximumn value of f can be obtained via $\frac{\partial f}{\partial\boldsymbol{\theta}_k[n]}=\mathbf{a}_k[n]-t_{2,k}^{-1}[n]\boldsymbol{\theta}_k[n]\overset{!}{=}0$, which leads to $\boldsymbol{\theta}_k[n]^\star=t_{2,k}[n]\mathbf{a}_k[n]$. Putting $\boldsymbol{\theta}_k[n]^\star$ back to (23) results in the same expression as in (18), which concludes the proof. □

Employing Lemma 1, we can transform the constraint (23) into the LMIs

$$\exists\boldsymbol{\theta}_k[n]\Big|\begin{bmatrix}t_{2,k}[n]\mathbf{I} & \boldsymbol{\theta}_k[n]\\ (\boldsymbol{\theta}_k[n])^{\mathrm{T}} & \gamma_k[n]\end{bmatrix}\succeq 0,\ \forall k\in\mathcal{K},n\in\mathcal{N}. \tag{24}$$

With the aforementioned linear matrix inequality transformations, we can rewrite the problem in (16) into an equivalent form as

$$\begin{aligned}\max_{\mathbb{P}_{\mathrm{G}},\mathbb{P}_{\mathrm{U}},\mathbb{Q},\mathbb{R},\mathbb{T}}\quad & (16a)\\ \text{s.t.}\quad & (16b),\ (21),\ (22),\ (24),\ (1),\ (4).\end{aligned} \tag{25}$$

Note that the problem in (25) is still non-convex due to the convex terms in the objective function (16a) and the concave terms in the constraints (16b). We apply the SIC method [9] to approximate the non-convex terms with the first order Taylor expansion of the corresponding terms and iteratively solve the problem. Concretely, in the ℓ-th iteration the problem in (25) can be approximated as

$$\begin{aligned}\max_{\mathbb{V}^{(\ell)}}\quad & \frac{1}{N}\sum_{n=1}^{N}\Big(\log_2\left(f_1\left(t_1^{(\ell)}[n]\right)+\sigma_{\mathrm{SI}}^2P_{\mathrm{U}}^{(\ell)}[n]+N_{\mathrm{U}}\right) + f_2\left(P_{\mathrm{U}}^{(\ell)}[n]\right)-(R_{\mathrm{E}}^{\max}[n])^{(\ell)}\Big)\\ \text{s.t.}\quad & (R_{\mathrm{E}}^{\max}[n])^{(\ell)}\geq f_3\left(P_{\mathrm{G}}^{(\ell)}[n],t_{2,k}^{(\ell)}[n]\right) -\log_2\left(f_4\left(t_{3,k}^{(\ell)}[n]\right)+N_{\mathrm{E},k}\right),\forall k\in\mathcal{K},n\in\mathcal{N},\\ & (21),\ (22),\ (24),\ (1),\ (4),\end{aligned} \tag{26}$$

where $\mathbb{V}^{(\ell)}:=\{\mathbb{P}_{\mathrm{G}}^{(\ell)},\mathbb{P}_{\mathrm{U}}^{(\ell)},\mathbb{Q}^{(\ell)},\mathbb{R}^{(\ell)},\mathbb{T}^{(\ell)}\}$ is the variable set, and $f_i(\cdot), i=1,2,3,4$ are the approximated terms given in (27). Note that problem in (26) is a joint convex problem that can be solved via convex solvers. The detailed algorithm is shown in Algorithm 1.

Algorithm 1 Iterative SIC based algorithm for optimal power-trajectory design under known Eves' locations

1: $\ell\leftarrow 0$; Set iteration number to zero
2: $\mathbb{P}_{\mathrm{G}}^{(0)},\mathbb{P}_{\mathrm{U}}^{(0)},\mathbb{T}^{(0)},\mathbb{Q}^{(0)}$; Initialize variables with feasible solution
3: **repeat**
4: $\quad\ell\leftarrow\ell+1$;
5: $\quad\boldsymbol{\theta}_k^{(\ell)}[n]\leftarrow t_{2,k}^{(\ell-1)}[n]\mathbf{a}_k^{(\ell-1)}[n]$;
6: $\quad\mathbb{P}_{\mathrm{G}}^{(\ell)},\mathbb{P}_{\mathrm{U}}^{(\ell)},\mathbb{Q}^{(\ell)},\mathbb{T}^{(\ell)}\leftarrow$ solve (26);
7: **until** a stable point or maximum number of ℓ reached
8: **return** $\left\{\mathbb{P}_{\mathrm{G}}^{(\ell)},\mathbb{P}_{\mathrm{U}}^{(\ell)},\mathbb{Q}^{(\ell)}\right\}$

IV. Robust Design based on Cutting Set Method

In the previous section, we proposed an algorithm for solving the problem with known Eves' locations. In this section, we cope with the uncertainty of the Eves' locations by applying the cutting-set method, which separates the worst-case optimization from the robust power and trajectory design. Specifically, the whole problem is solved through alternating between two steps, i.e., power-trajectory design and worst-case channel determination. In the first step, i.e., the step of power-trajectory design, the transmit power of GU and UAV as well as the trajectory of UAV are optimally obtained under

$$\left(t_1^{(\ell)}[n]\right)^{-1} \geq -\left(t_1^{(\ell-1)}[n]\right)^{-2}\left(t_1^{(\ell)}[n]-t_1^{(\ell-1)}[n]\right)+\left(t_1^{(\ell-1)}[n]\right)^{-1} \triangleq f_1\left(t_1^{(\ell)}[n]\right), \tag{27a}$$

$$\left(t_{3,k}^{(\ell)}[n]\right)^{-1} \geq -\left(t_{3,k}^{(\ell-1)}[n]\right)^{-2}\left(t_{3,k}^{(\ell)}[n]-t_{3,k}^{(\ell-1)}[n]\right)+\left(t_{3,k}^{(\ell-1)}[n]\right)^{-1} \triangleq f_4\left(t_{3,k}^{(\ell)}[n]\right), \tag{27b}$$

$$-\log_2\left(\sigma_{\mathrm{SI}}^2 P_{\mathrm{U}}^{(\ell)}[n]+N_{\mathrm{U}}\right) \geq \frac{-\sigma_{\mathrm{SI}}^2\left(P_{\mathrm{U}}^{(\ell)}[n]-P_{\mathrm{U}}^{(\ell-1)}[n]\right)}{\ln(2)\left(\sigma_{\mathrm{SI}}^2 P_{\mathrm{U}}^{(\ell-1)}[n]+N_{\mathrm{U}}\right)}-\log_2\left(\sigma_{\mathrm{SI}}^2 P_{\mathrm{U}}^{(\ell-1)}[n]+N_{\mathrm{U}}\right) \triangleq f_2\left(P_{\mathrm{U}}^{(\ell)}[n]\right), \tag{27c}$$

$$\begin{aligned}\log_2\left(\frac{\beta_0 P_{\mathrm{G}}^{(\ell)}[n]}{(\mathbf{c}_k^{\mathrm{T}}\mathbf{c}_k)^{\frac{\alpha}{2}}}+t_{2,k}^{(\ell)}[n]+N_{\mathrm{E},k}\right) &\leq \frac{\frac{\beta_0\left(P_{\mathrm{G}}^{(\ell)}[n]-P_{\mathrm{G}}^{(\ell-1)}[n]\right)}{(\mathbf{c}_k^{\mathrm{T}}\mathbf{c}_k)^{\frac{\alpha}{2}}}+t_{2,k}^{(\ell)}[n]-t_{2,k}^{(\ell-1)}[n]}{\ln(2)\left(\frac{\beta_0 P_{\mathrm{G}}^{(\ell-1)}[n]}{(\mathbf{c}_k^{\mathrm{T}}\mathbf{c}_k)^{\frac{\alpha}{2}}}+t_{2,k}^{(\ell-1)}[n]+N_{\mathrm{E},k}\right)}+\log_2\left(\frac{\beta_0 P_{\mathrm{G}}^{(\ell-1)}[n]}{(\mathbf{c}_k^{\mathrm{T}}\mathbf{c}_k)^{\frac{\alpha}{2}}}+t_{2,k}^{(\ell-1)}[n]+N_{\mathrm{E},k}\right)\\ &\triangleq f_3\left(P_{\mathrm{G}}^{(\ell)}[n], t_{2,k}^{(\ell)}[n]\right).\end{aligned} \tag{27d}$$

the assumption that the Eves' locations belong to a certain finite subset of uncertainty region. Then, the second step, i.e., the step of worst-case channel determination finds the Eves' locations corresponding to the worst-case under the power-trajectory design obtained in the first step. The details of each step are given in the following.

A. Power-Trajectory Design under Finite Subsets of Eves' Locations

In this step, the worst-case location of Eves at the n-th time slot is assumed to be included in a finite subset of potential Eves' locations[2]. Specifically, let $\mathcal{C}_k = \{\hat{\mathbf{c}}_k + \Delta\mathbf{c}_k | \Delta\mathbf{c}_k \in \Omega_k\}$ be the potential locations of the k-th Eve. Then, the finite subset of the k-th Eve's location at the n-th time slot is denoted as $\tilde{\mathcal{C}}_k[n] \subset \mathcal{C}_k$. Note that with these subsets of Eves' locations it is actually to consider $\tilde{K}[n]$ number of Eves at n-th time slot, where $\tilde{K}[n] = |\tilde{\mathcal{C}}[n]|$ with $\tilde{\mathcal{C}}[n] = \tilde{\mathcal{C}}_1[n] \cup \tilde{\mathcal{C}}_2[n] \cup \cdots \cup \tilde{\mathcal{C}}_K[n]$ and $|\mathcal{A}|$ is the cardinality of set $\mathcal{A}$. Then, the problem is transformed into the optimal power-trajectory design problem under known Eves' locations as

$$\begin{aligned} \max_{\mathbb{P}_{\mathrm{G}},\mathbb{P}_{\mathrm{U}},\mathbb{Q},\mathbb{R}} \quad & \frac{1}{N}\sum_{n=1}^{N}\left(\check{R}_{\mathrm{U}}[n]-R_{\mathrm{E}}^{\max}[n]\right) \\ \text{s.t.} \quad & R_{\mathrm{E}}^{\max}[n] \geq \hat{R}_{\mathrm{E},k}[n],\ \forall k \in \tilde{\mathcal{K}}[n], n \in \mathcal{N} \\ & (1),\ (4), \end{aligned} \tag{28}$$

where $\tilde{\mathcal{K}}[n] = \{1,\ldots,\tilde{K}[n]\}$ is the index set of the considered Eves at n-th time slot. Note that problem in (28) can be solved using the Algorithm 1 elaborated in Section III.

B. Worst-Case Eve's Location Determination for Given Power-Trajectory Design

In this step, the worst-case location of Eve at each time slot is found under the given power-trajectory design obtained from the first step. We first discretize the potential locations of the k-th Eve $\mathcal{C}_k$ into the equal spacing grid based searching points with the resolution of ξ, which is denoted as $\bar{\mathcal{C}}_k \subset \mathcal{C}_k$. Moreover, for the simplification of notation we define the number of total searching points as $\bar{K} = |\bar{\mathcal{C}}|$, where $\bar{\mathcal{C}} = \bar{\mathcal{C}}_1 \cup \bar{\mathcal{C}}_2 \cup \cdots \cup \bar{\mathcal{C}}_K$. Then, we find the locations that correspond to the worst average secrecy rate by searching through the discrete points in the set of $\bar{\mathcal{C}}$. Note that the case of the worst average secrecy rate corresponds to the case where the system has the worst secrecy rate at each time slot. Thus, the searching process can be applied separately for each time slot. In particular, the worst-case Eve's location at the n-th time slot, denoted as $\bar{\mathbf{c}}[n]$, is obtained as

$$\bar{\mathbf{c}}[n] = \arg\max_{\mathbf{c}[n]\in\bar{\mathcal{C}}} \hat{R}_{\mathrm{E},k}[n],\ \forall n \in \mathcal{N}, \tag{29}$$

where $\hat{R}_{\mathrm{E},k}[n]$ can be directly calculated with the given power-trajectory design. Finally, the obtained worst-case Eves' locations are added into the finite subset of potential Eves' locations $\tilde{\mathcal{C}}[n]$ for the next turn of power-trajectory design.

C. Total Algorithm for Robust Power-Trajectory Design

As aforementioned the cutting-set algorithm solves the robust power-trajectory optimization problem alternately through two previously elaborated steps. In this part, we provide the total iterative algorithm. At the beginning, the initial subsets of potential Eves' locations are given as $\{\tilde{\mathcal{C}}^{(1)}[n]\}$. At the i-th iteration the optimal power-trajectory design is calculated under the finite subsets of Eves' locations $\{\tilde{\mathcal{C}}^{(i)}[n]\}$ by solving the problem in (28) utilizing the method given in Algorithm 1. Then, by searching the grids the worst-case Eves' locations are obtained following (29) as $\{\bar{\mathbf{c}}^{(i)}[n]\}$ and they are added into the finite subsets for the next iteration as $\tilde{\mathcal{C}}^{(i+1)}[n] = \{\tilde{\mathcal{C}}^{(i)}[n], \bar{\mathbf{c}}^{(i)}[n]\}, \forall n \in \mathcal{N}$. These two steps are alternatively processed until the improvement between two iterations is less than the threshold or a maximum iteration number is reached. The whole algorithm is detailed shown in Algorithm 2.

Algorithm 2 Cutting-set method based algorithm for power-trajectory design with uncertainty of Eves' locations

```
1: i ← 1;   Set iteration number of cutting-set method to one
2: C̃^(1)[n];   Initialize finite subsets
3: repeat
4:     P_G^(i), P_U^(i), Q^(i) ← solve (28) under {C̃^(i)[n]} using Algorithm 1;
5:     {c̄^(i)[n]} ← (29) based on P_G^(i), P_U^(i), Q^(i);
6:     C̃^(i+1)[n] ← {C̃^(i)[n], c̄^(i)[n]}, ∀n ∈ N;   Update finite subsets
7:     i ← i + 1;
8: until a stable point or maximum number of i reached
9: return {P_G^(i), P_U^(i), Q^(i)}
```

[2]Note that we consider the worst case where Eves' locations at each time slot are independent from other time slots, while the uncertainty region of each Eve is fixed among all time slots as $\Omega_k, \forall k \in \mathcal{K}$.

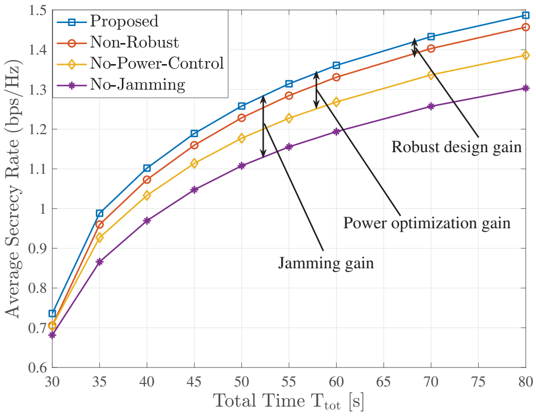

Fig. 2. Average secrecy rate vs. total flying time T_{tot}.

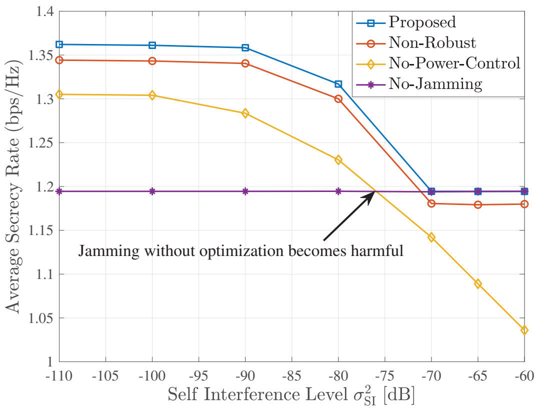

Fig. 3. Average secrecy rate vs. self interference level σ_{SI}^2.

V. Numerical Results

In this section we numerically evaluate the proposed algorithm via the comparison with the following benchmark schemes. 1) Non-Robust: this scheme solves the problem via Algorithm 1 under the estimated Eves' locations. 2) No-Power-Control: this scheme allocates the average power along the whole time, i.e., $P_{\mathrm{G}}[n] = P_{\mathrm{G}}^{\mathrm{avg}}, P_{\mathrm{U}}[n] = P_{\mathrm{U}}^{\mathrm{avg}}, \forall n \in \mathcal{N}$. 3) No-Jamming: this scheme provides a robust design via Algorithm 2 without jamming, i.e., $P_{\mathrm{U}}[n] = 0, \forall n \in \mathcal{N}$. The simulation parameters are set as: $K = 2$ Eves with estimated locations $\hat{\mathbf{c}}_1 = (-100, -50, 0)$m, $\hat{\mathbf{c}}_2 = (150, -100, 0)$m and estimation errors $\epsilon_1 = 50$m, $\epsilon_2 = 100$m. $\alpha = 3$, $\xi = 1$m, $\mathbf{q}[0] = (-400, -200, 100)$m, $\mathbf{q}[N+1] = (200, -200, 100)$m, $z_{\min} = 90$m, $z_{\max} = 110$m, $V_{\max} = 20$m/s, $V_{\mathrm{up}} = V_{\mathrm{down}} = 5$m/s, $P_{\mathrm{G}}^{\mathrm{avg}} = P_{\mathrm{U}}^{\mathrm{avg}} = 20$dBm, $P_{\mathrm{G}}^{\mathrm{peak}} = P_{\mathrm{U}}^{\mathrm{peak}} = 26$dBm, $\delta = 0.5$s, $N_{\mathrm{U}} = N_{\mathrm{E},k} = N_0$ and $\beta_0/N_0 = 60$dB.

In Fig. 2, the average secrecy rates in bits/second/Hertz (bps/Hz) with $\sigma_{\mathrm{SI}}^2 = -90$dB under various total flight time T_{tot} are depicted. It is observed that the resulting secrecy rates of all algorithms increase significantly with T_{tot}. This is because higher total flying time leads a longer hovering time of the UAV around the location of the GU. A significant performance gain is shown due to jamming and optimized power control under a wide range of T_{tot}. Furthermore, the improvement of the proposed algorithm compared to other benchmarks is more significant with a larger T_{tot}, since a large T_{tot} provides more flexibility for power control and trajectory design under the awareness of estimation error of Eves' locations.

In Fig. 3, the impact of loop interference level is evaluated under a total flying time $T_{\mathrm{tot}} = 60$s. It is observed that under weak SI, i.e., small σ_{SI}^2, jamming gain is significant, while under strong SI, i.e., large σ_{SI}^2, the results of the proposed algorithm converge to No-Jamming design. This shows that when σ_{SI}^2 is large the capability of SI cancellation at the UAV is declined and thus the optimal jamming power would be zero to avoid the performance decrease due to strong SI. Oppositely, the performance of No-Power-Control scheme degrades dramatically with the increase of σ_{SI}^2 because of the fixed power allocation strategy.

VI. Conclusion

This paper investigated a secure FD-jamming enabled UAV communication system with the consideration of the uncertainty of multiple Eves' locations. We proposed a robust transmit power and trajectory design to maximize the worst average secrecy rate under the UAV's mobility constraints and the power budget limitations. The numerical results show that under a weak residual SI level FD jamming can significantly enhance the system performance in terms of average worst secrecy rate.

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